Visualizing spatio-temporal distribution of flow data

Jacques Gautier, Vincent HEAU, Jean-Baptiste BESNIER, Corentin PEUTIN et Lorenzo LOPEZ UROS
IGN-ENSG, LaSTIG, Equipe GeoVIS
11/03/2025

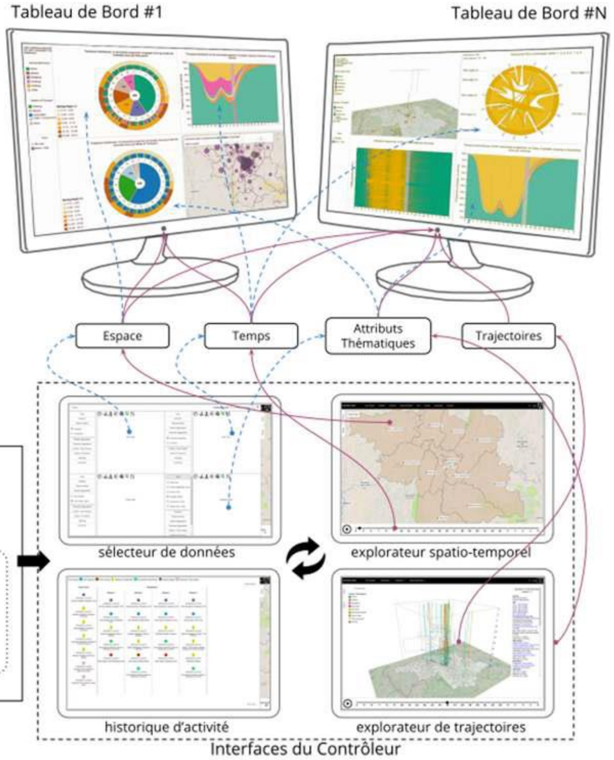
How to visualy explore origin-destination flow data?

Exploring flow data distribution according to their spatial or temporal components

Identifying correlation between their spatial and temporal distribution

Through different spatial and temporal scales

Visual exploration of flow data



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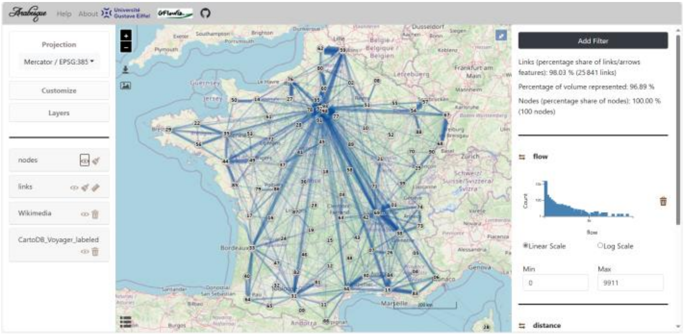
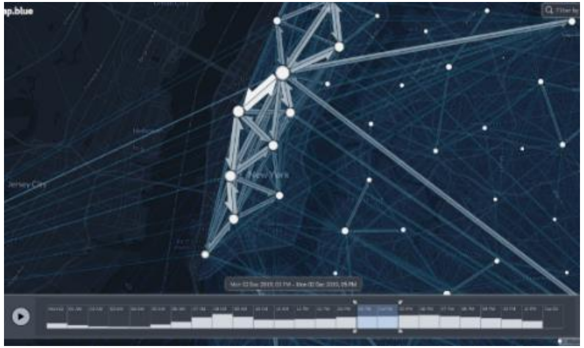


Roelandt, N., Bahoken, F., Le Campion, G., Jegou, L., Maisonobe, M., Mericskay, B., & Come, E. (2021). One Arabesque in the small world of OD webmaps. ISPRS International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 46, 147-154.



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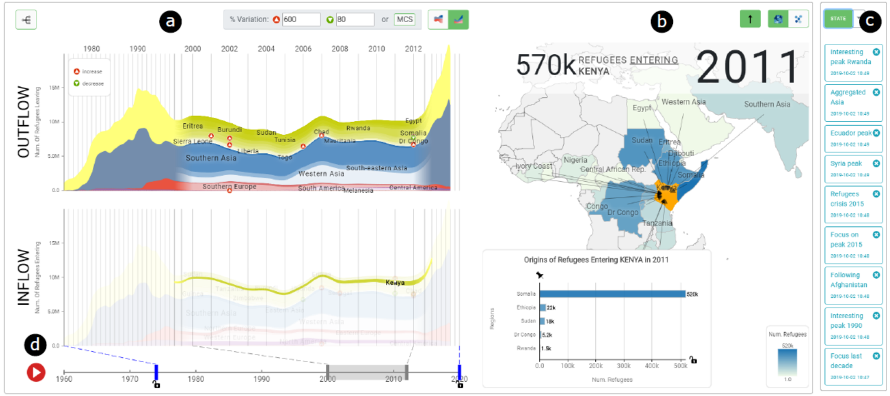
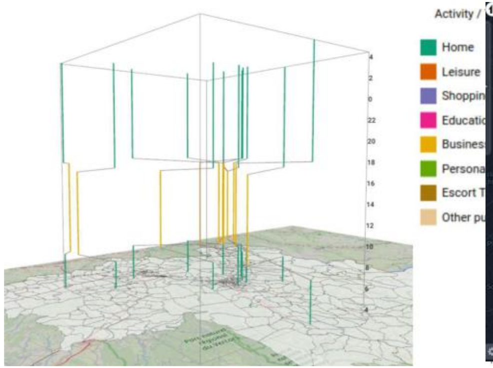
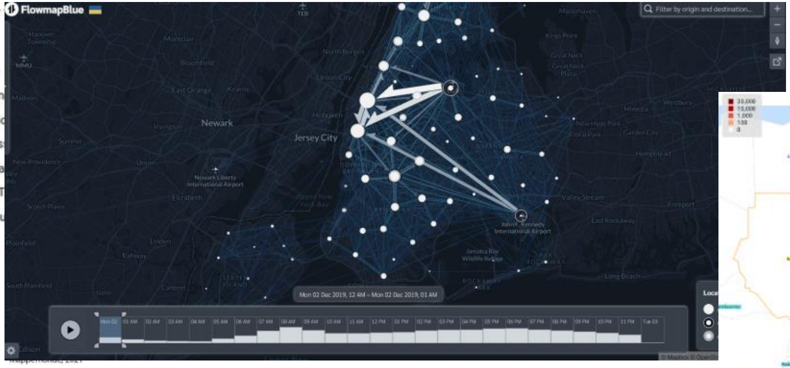


Cuenca, E., éd éric Docquier, F., Nijssen, S., & Schaus, P. EvoFlows: an Interactive Approach for Visualizing Spatial and Temporal Trends in Origin-Destination Data.

Integrating spatial and temporal component into the same graphical space





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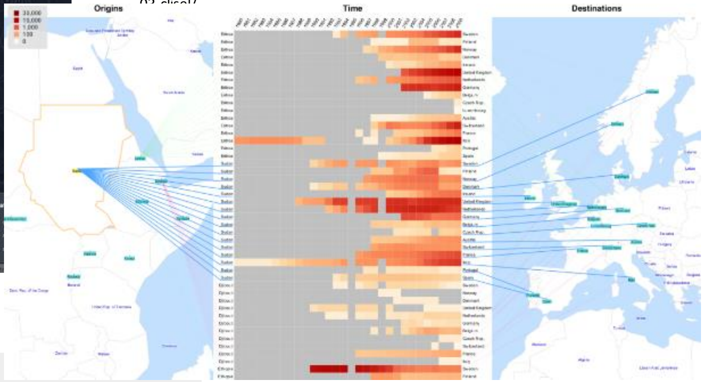

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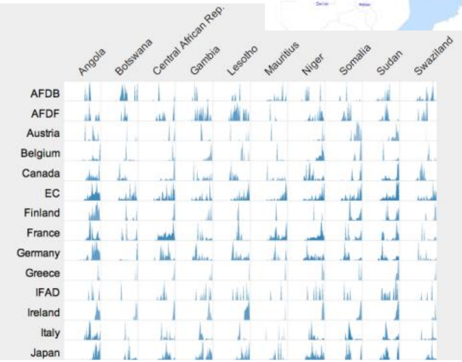
Boyandin, I., Bertini, E., & Lalanne, D. (2010, May). Using flow maps to explore migrations over time. In *Geospatial visual analytics workshop in conjunction with the 13th AGILE international conference on geographic information science* (Vol. 2, No. 3).



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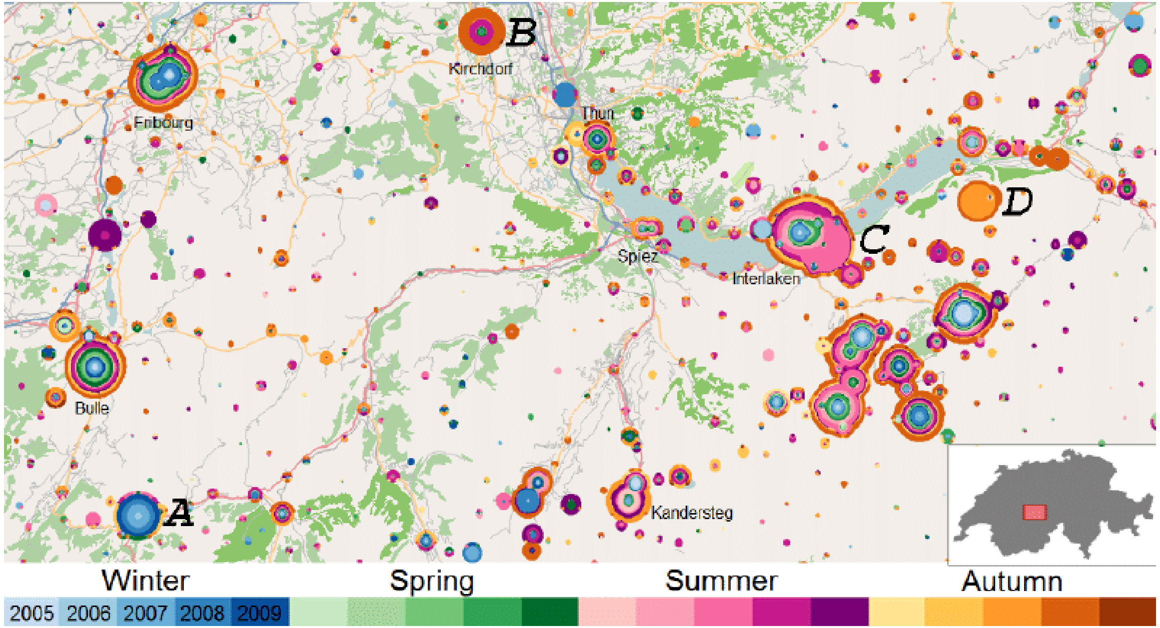
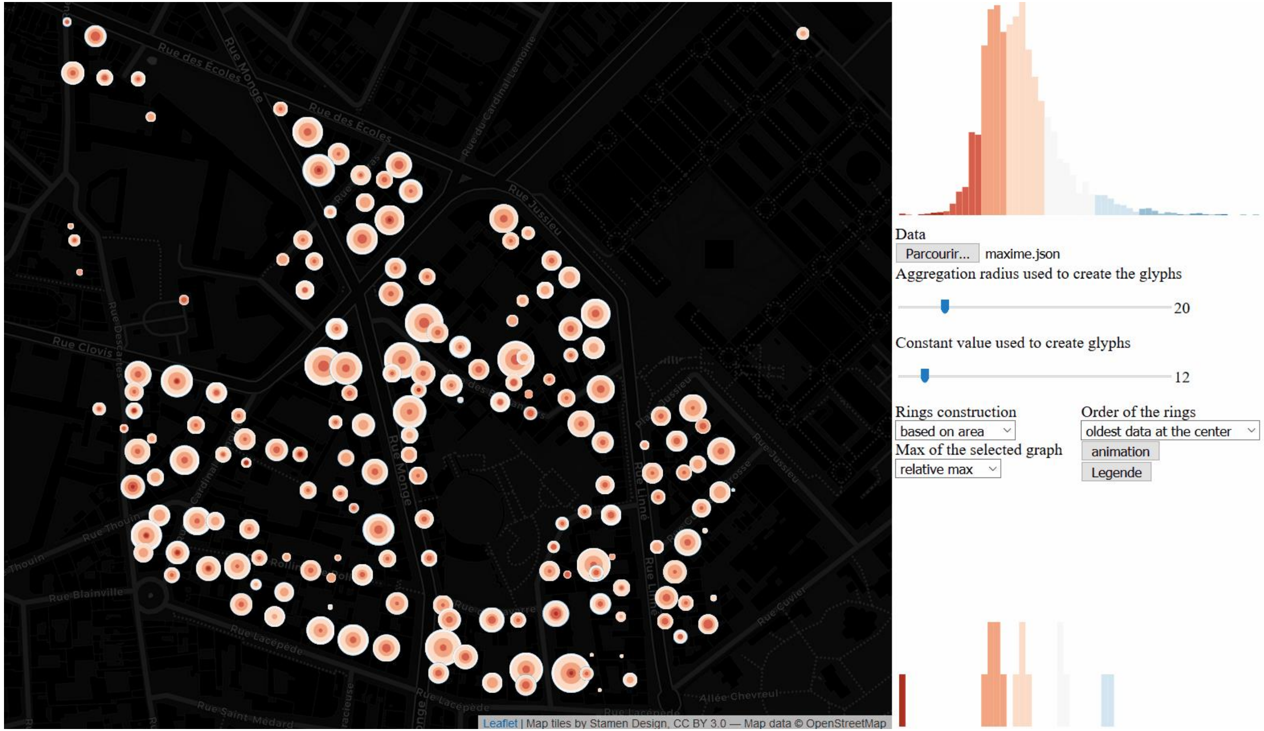


Growth Ring Maps

Bak, P., Mansmann, F., Janetzko, H., & Keim, D. (2009). Spatiotemporal analysis of sensor logs using growth ring maps. *IEEE transactions on visualization and computer graphics*, *15*(6), 913-920.

Representing, through color, of the temporal distribution of spatially aggregated events

Integrating spatial and temporal component into the same graphical space



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Integration in a multiscale interactive visulization

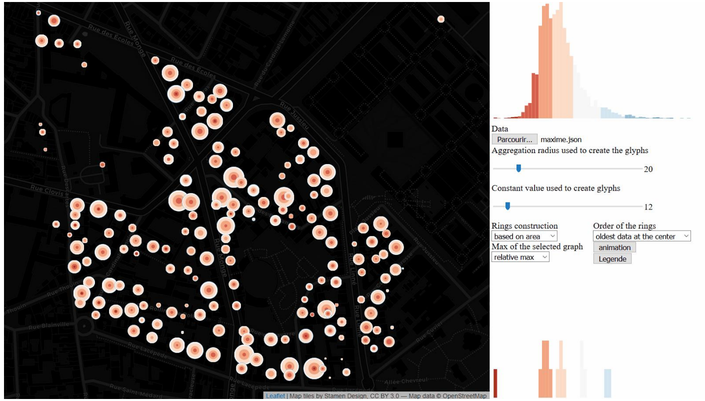
Overview of the events spatio-temporel distribution

- Clusters identification
- Local trends identification

Exploration through space and time with interaction

- Spatial zoom / selection
- Modification of color scale, temporal selection

Combining the GRM approach with a multiscale interactive flow map



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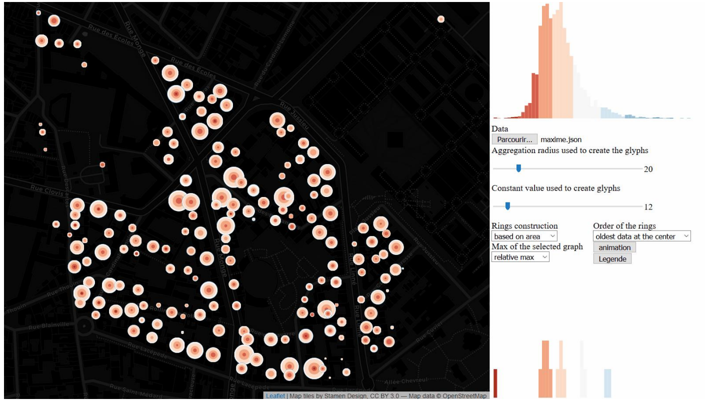

Encoding temporal distribution of OD flow data...

...through the symbols representing the links and nodes of the OD network

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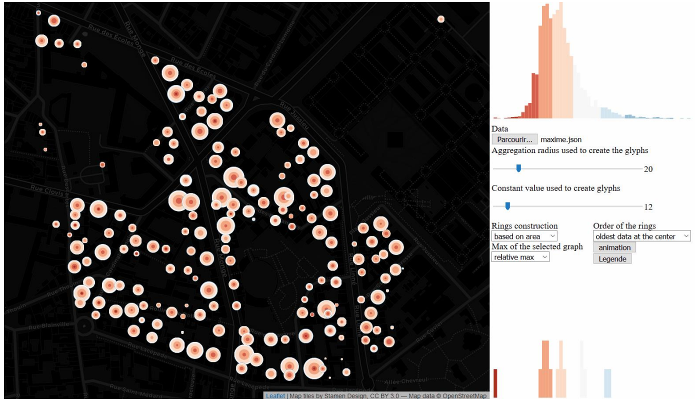

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- Visual exploration through ≠ spatial and temporal scales
- Vue d'ensemble de la distribution ST des flux
- Comparing flow temporal distribution at several spatial locations

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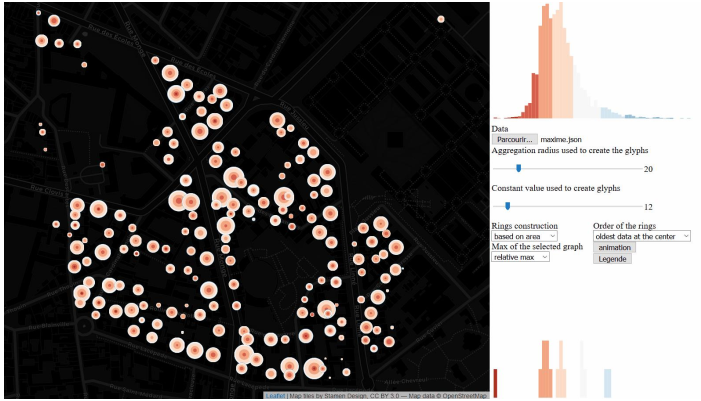

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Application to urban mobility analysis

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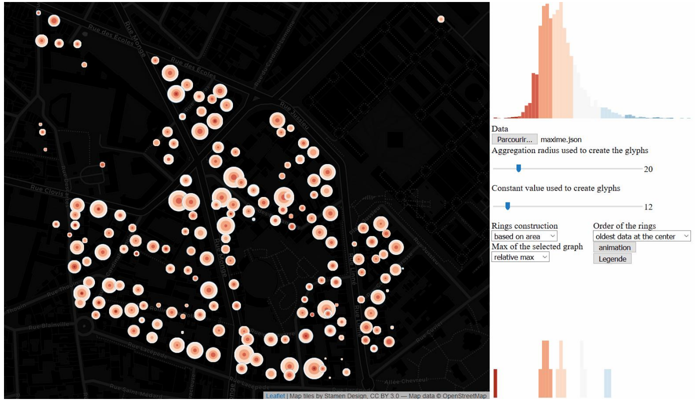

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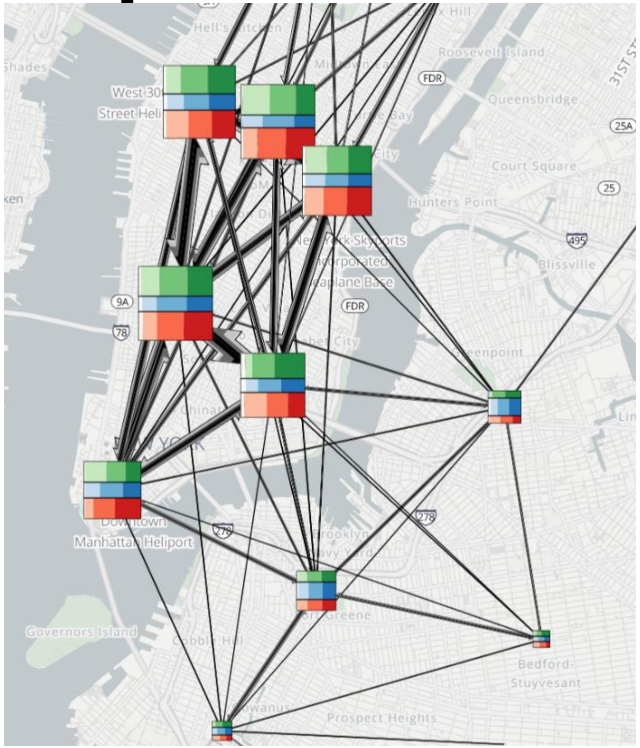

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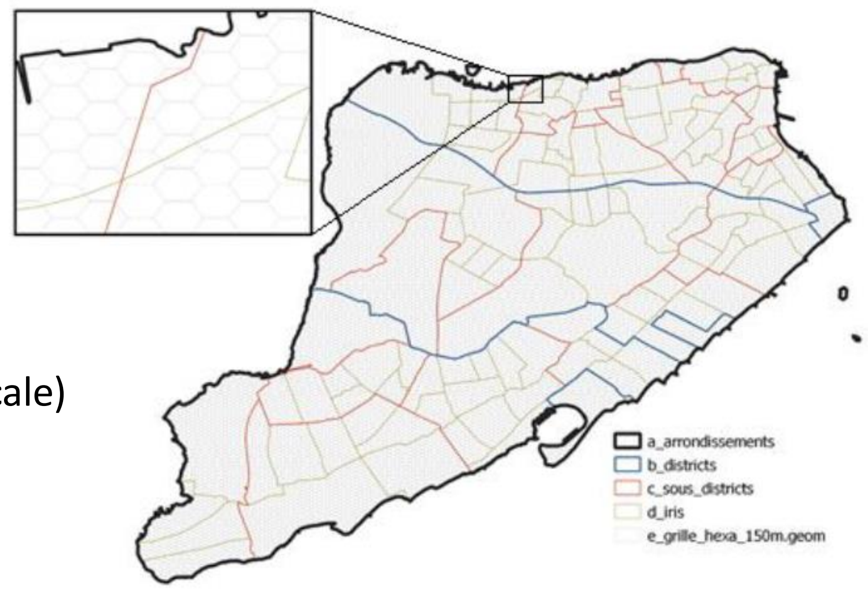
Design creation

Individual travels data (without trajectory)

Travels between bycicle sharing system stations in NY

- Origin-destination coordinates
- Time of departure/arrival

- Spatial aggregation of the origin/destination points through different grids (according to scale)



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Individual travels data (without trajectory)

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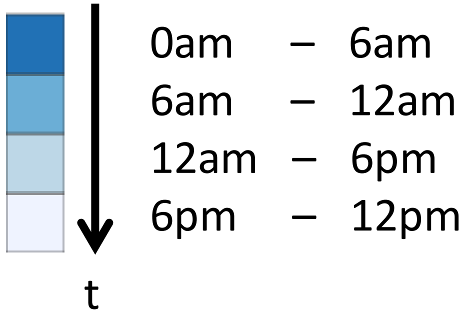
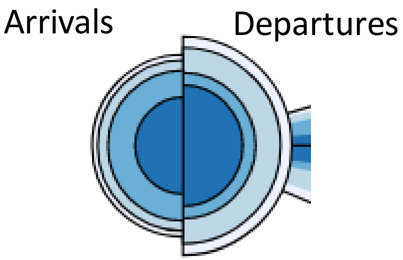
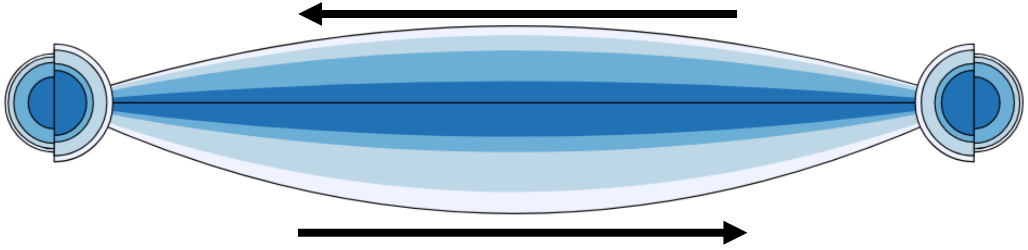
- Spatial aggregation of the origin/destination points through different grids (according to scale)
- Encoding number of travels with symbol's size
- Encoding departure time with symbol's color







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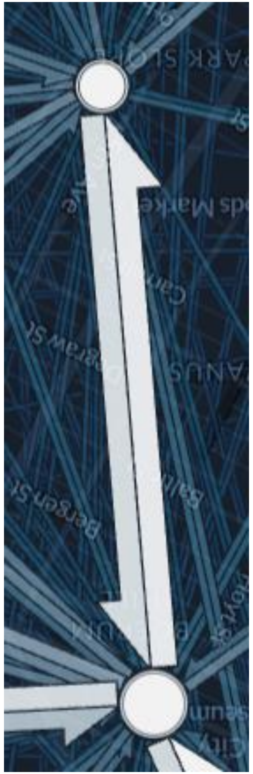

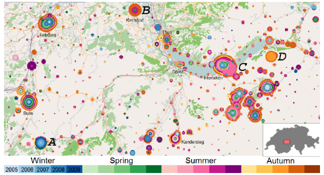

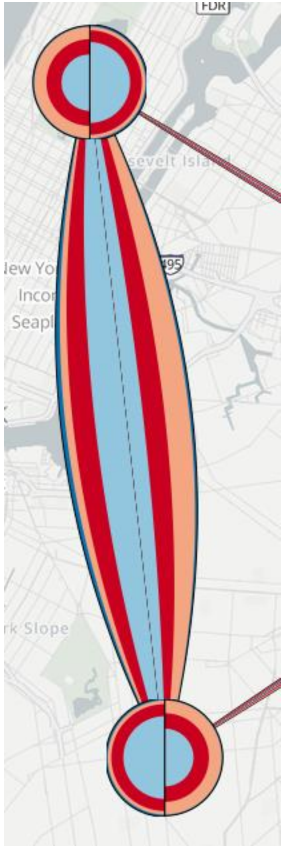

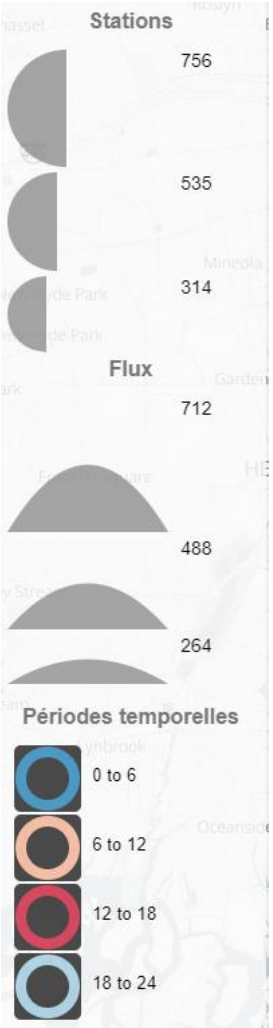

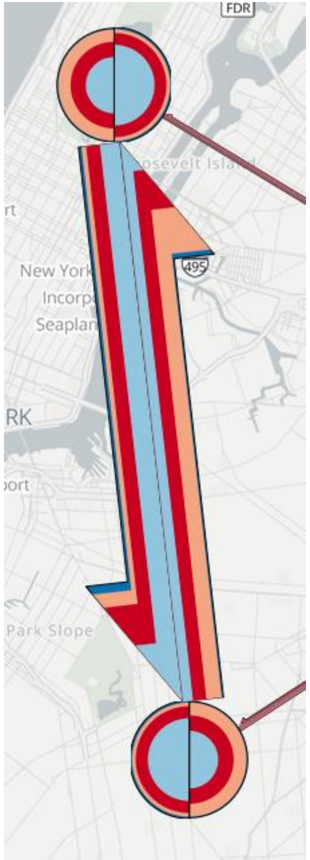

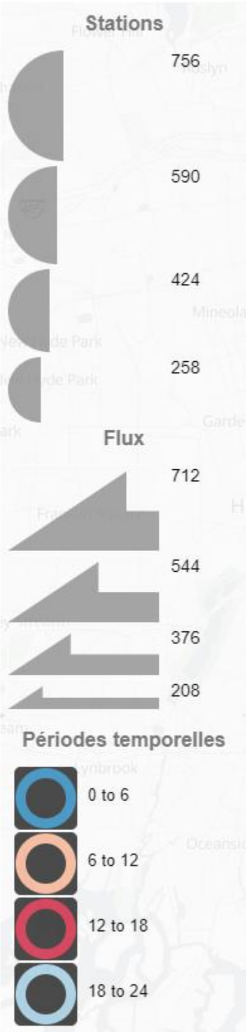

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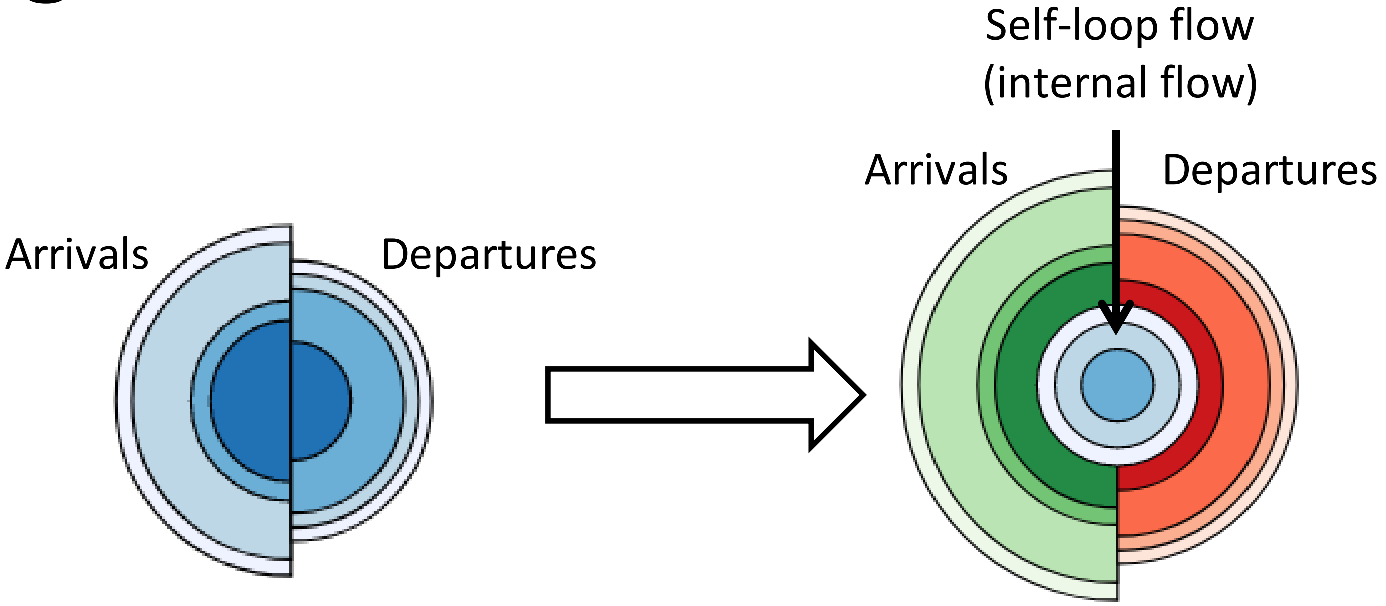
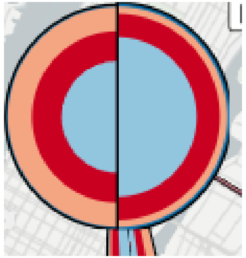


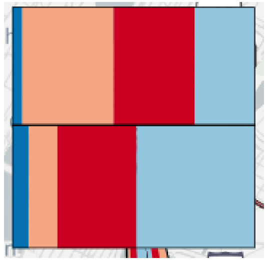
Design creation

Arrivals



Departures

Arrivals



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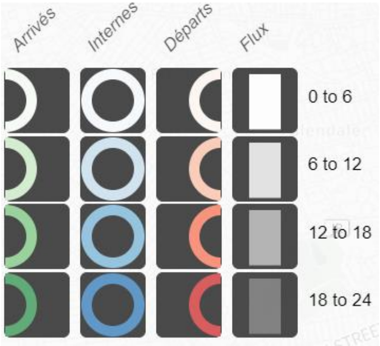
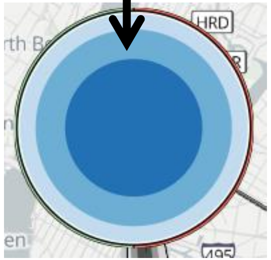


Self-loop flow
(internal flow)

Arrivals

Departures



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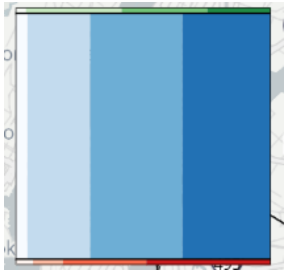

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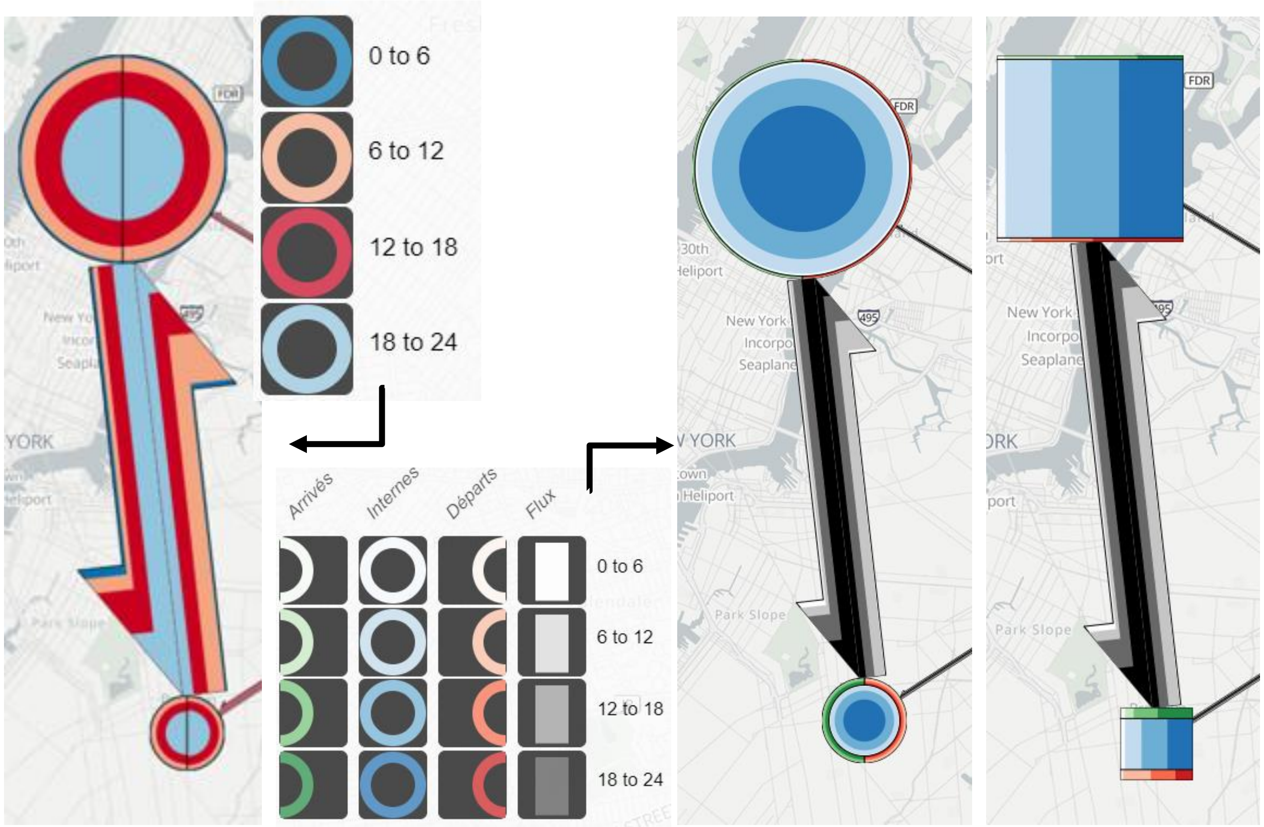
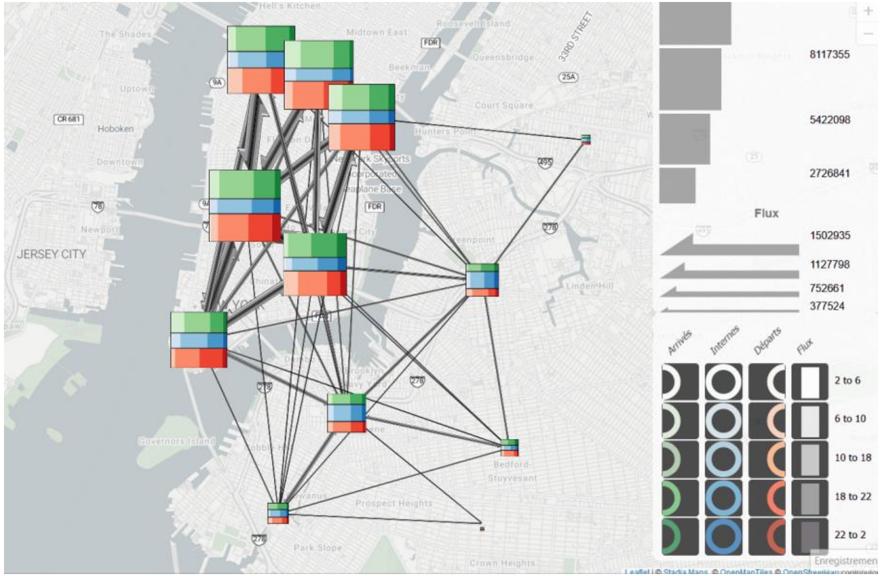


Exploring S and T components

By interacting with the visualization

- **Selection of areas to highlight**
  Click on one or several nodes, displaying only data related to the selected nodes
- **Changing the spatial scale**
  Zooming and changing spatial aggregation granularity



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- **Selection of areas to highlight**
  Click on one or several nodes, displaying only data related to the selected nodes
- **Changing the spatial scale**
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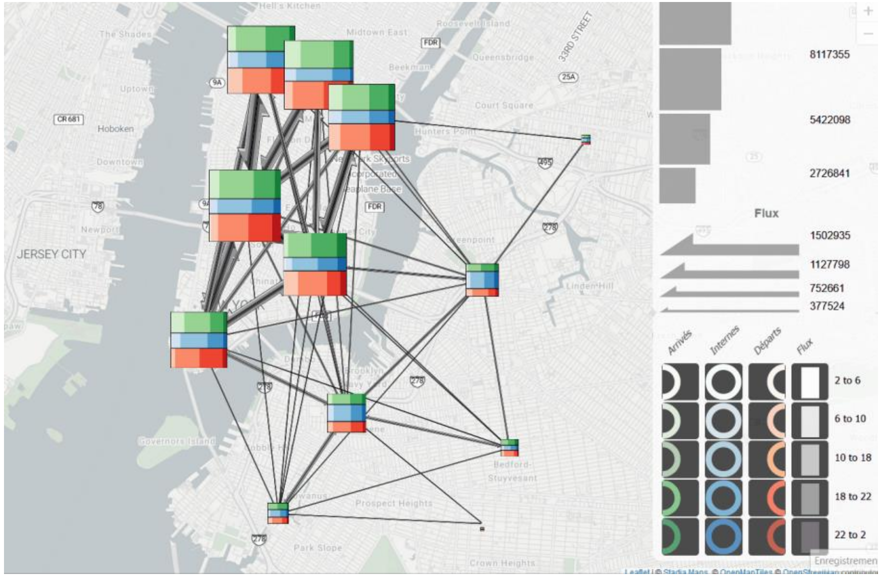
- **Selection of periods to highlight**
  Click on one or several temporal periods in the legend, displaying only data related to the selected periods
- **Changing the temporal scale**
  - Changing the temporal extent
  - Changing the temporal classification (number of classes, linear/cyclic classification)

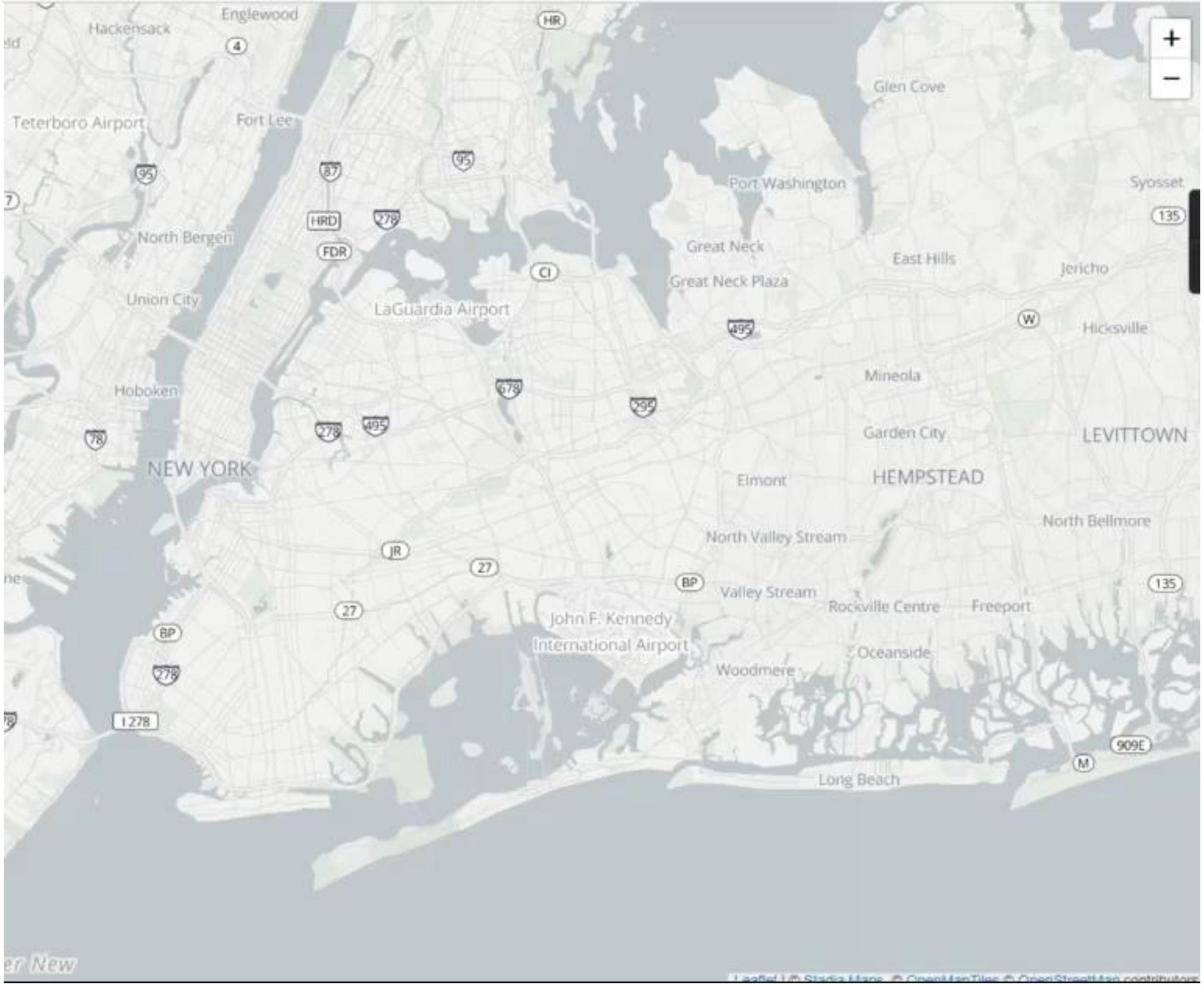
  Choice between several datasets



Unintuitive modalities of interaction with data temporal component

Results

- Data overview
- Zooming on an area with a lot of internal flow
- Investigating flow related to a specific location
- Comparing travels to/from this point at specific periods
- …

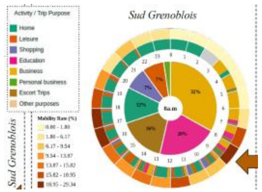


And now

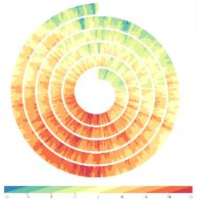
- *Submitted to SAGEO 2025*
- *Integration into GEOVIS future works in PEPR Mobidec (MobSci Data Factory) with Maria Jesus Lobo*

Futur works

- *Which interfaces for a better interaction with temporal component ?*



esTime, Menin, 2020



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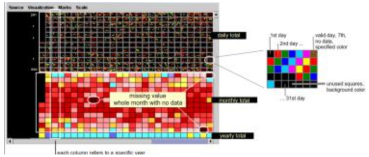

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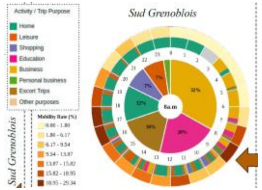

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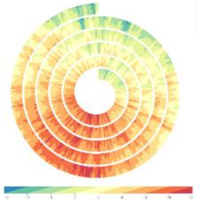

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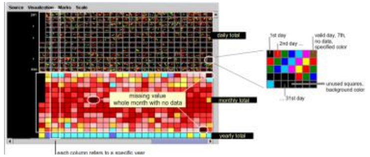

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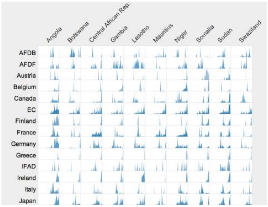

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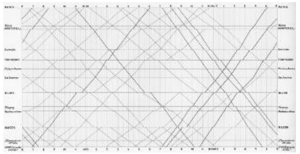

Marey's train schedule (1885)
Tufte (1990)

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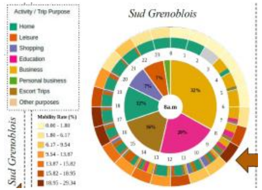
- *Application to trajectory data ?*



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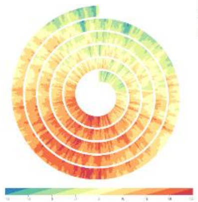

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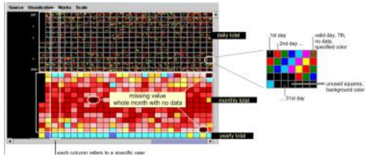

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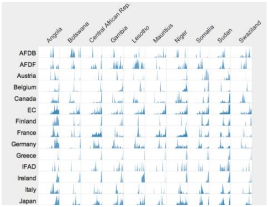

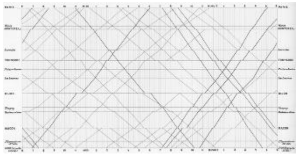

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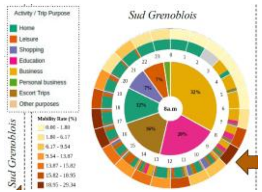
- *Application to trajectory data ?*
- *Intuitiveness/level of detail balance depending on use?*



esTime, Menin, 2020

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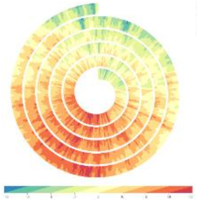

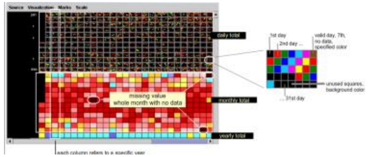

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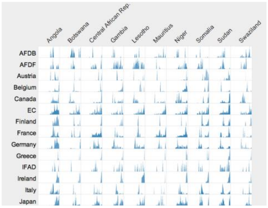

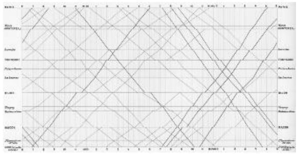

Marey's train schedule (1885)
Tufte (1990)

Thank you

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